Probability

This Chapter Overview

This chapter is a recap of the concepts you learnt at GCSE.

1: Basic Probability

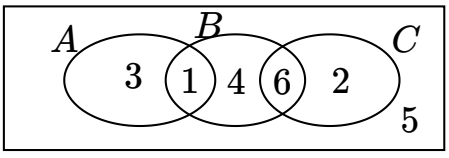
"I throw two fair die. Calculate the probability the sum of the two dice is more than 6."

2: Venn Diagrams

"Out of 50 students, 12 play both piano and drums, 30 play piano and 25 play drums. Find the probability a randomly chosen student plays neither instrument."

3: Mutually Exclusive/Independent Events

Determine whether A and B are independent.



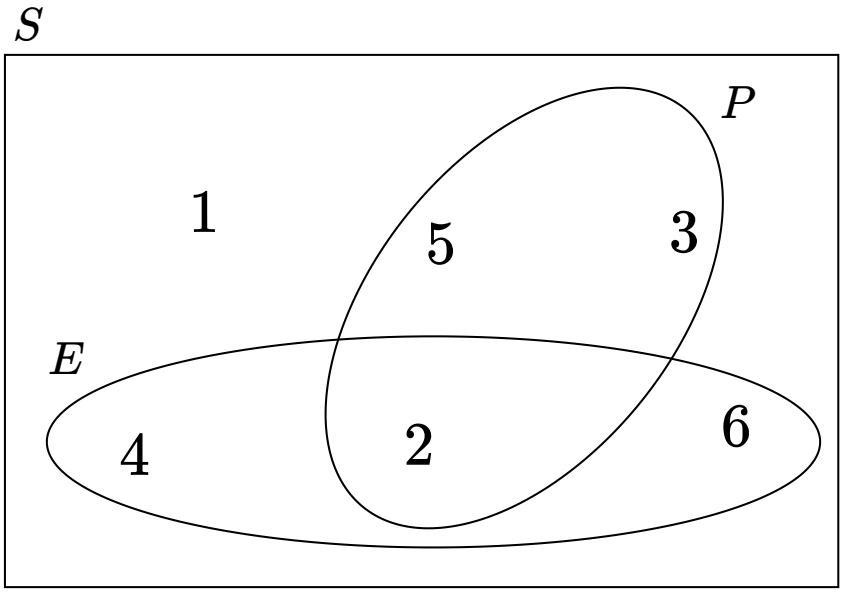
4: Tree Diagrams

"The probability I hit a target is 0.3. If I hit it, the probability I hit again on the next shot is 0.4. If I miss, the probability I hit on the next shot is 0.1. If I shoot 3 times, what's the probability I hit on the first and third shot?"



An **experiment** is a repeatable process that gives rise a number a number of **outcomes**.

An **event** is a set of one or more of these outcomes.
(We often use capital letters to represent them)



E = "rolling an even number"
P = "rolling a prime number"

A **sample space** is the set of all possible outcomes.

Because we are dealing with sets, we can use a **Venn diagram**, where
- the numbers are the individual outcomes,
- the sample space is a rectangle and
- the events are sets, each a subset of the sample space.

You do not need to use set notation like $\cap$ and $\cup$ in this module (but ordinarily you would!)

Two fair spinners each have four sectors numbered 1 to 4. The two spinners are spun together and the sum of the numbers indicated on each spinner is recorded.
Find the probability of the spinners indicating a sum of
(a) exactly 5 (b) more than 5

Spinner 2 \ Spinner 1	1	2	3	4
1	2	3	4	5
2	3	4	5	6
3	4	5	6	7
4	5	6	7	8

$P(\text{sum} = 5) =$ a

$P(\text{sum} > 5) =$ b

If the sample space is the amalgamation of two underlying experiments, a table is a helpful way to list the outcomes.

The table shows the times taken, in minutes, for a group of students to complete a number puzzle.

Time, t (min)	$5 \leq t < 7$	$7 \leq t < 9$	$9 \leq t < 11$	$11 \leq t < 13$	$13 \leq t < 15$
Frequency	6	13	12	5	4

A student is chosen at random. Find the probability for a group of students to complete a number puzzle

(a) In under 9 minutes (b) in over 10.5 minutes.

$P(t < 9) =$ a

$P(t > 10.5) =$ b

You're effectively doing **linear interpolation** here: 10.50 to 11 is a quarter of the $9 \leq t < 11$ interval, therefore we use a quarter of the 12 people.
This is an estimate because we're assuming the people are equally distributed across the interval.

3. The masses of 140 adult Bullmastiffs are recorded in a table. One dog is chosen at random.

(a) Find the probability that its mass is at least 54 kg.

(b) Find the probability that its mass is between 48 and 57 kg.

(c) A randomly chosen Rottweiler has probability 0.54 of weighing under 53 kg. Is a Bullmastiff more or less likely to weigh under 53 kg? State your assumption. Hint: use interpolation.

a

b

c

Mass (kg)	Frequency
$45 \leq m < 48$	17
$48 \leq m < 51$	25
$51 \leq m < 54$	42
$54 \leq m < 57$	33
$57 \leq m < 60$	21
$60 \leq m < 63$	2

4. The lengths of 240 koalas were recorded. One koala is chosen at random.

Find the probability that it is:

(a) female;

(b) less than 80 cm long;

(c) male, between 75 cm and 85 cm long.

(d) Koalas under 72 cm are juvenile. Estimate the probability of choosing a juvenile and state one assumption.

Length (cm)	Male	Female
$65 \leq l < 70$	4	14
$70 \leq l < 75$	20	15
$75 \leq l < 80$	24	32
$80 \leq l < 85$	47	27
$85 \leq l < 90$	31	26

a

b

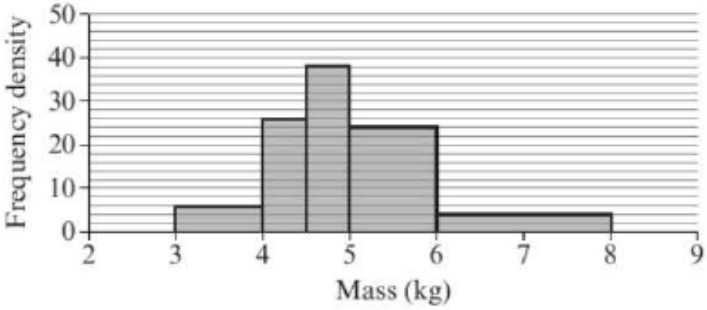
c

d

5. The histogram shows the distribution of masses of 70 adult cats.

(a) Find the probability that a randomly chosen cat has mass greater than 5 kg. (2 marks)

(b) Estimate the probability that a randomly chosen cat has mass less than 6.5 kg. (3 marks)



a

b

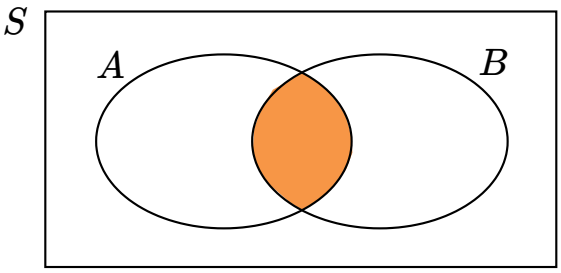
Extension 9

[Challenge p72] Samira chooses one card at random from each group. Group A has 2, 7, 5; group B has 4 and x. She records their product Y. Given x is an integer and $P(Y \text{ is even}) = P(Y \geq 20)$, find the possible values of x.

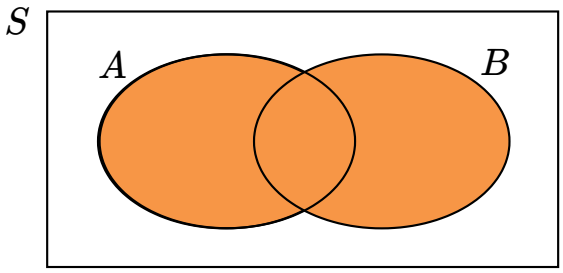
Solution

Venn Diagrams 10

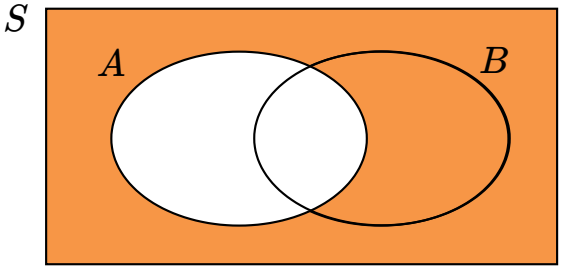
Venn Diagrams allow us to combine events, e.g. "A happened **and** B happened".



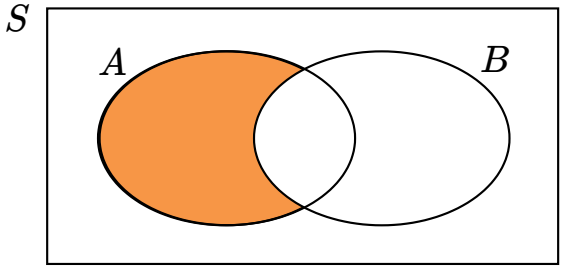
The event "A **and** B"
Known as the **intersection** of A and B.



The event "A **or** B"
Known as the **union** of A and B.



The event "not A"
Known as the complement of A.



These can be combined,
e.g. "A and not B".

Example involving probabilities 11

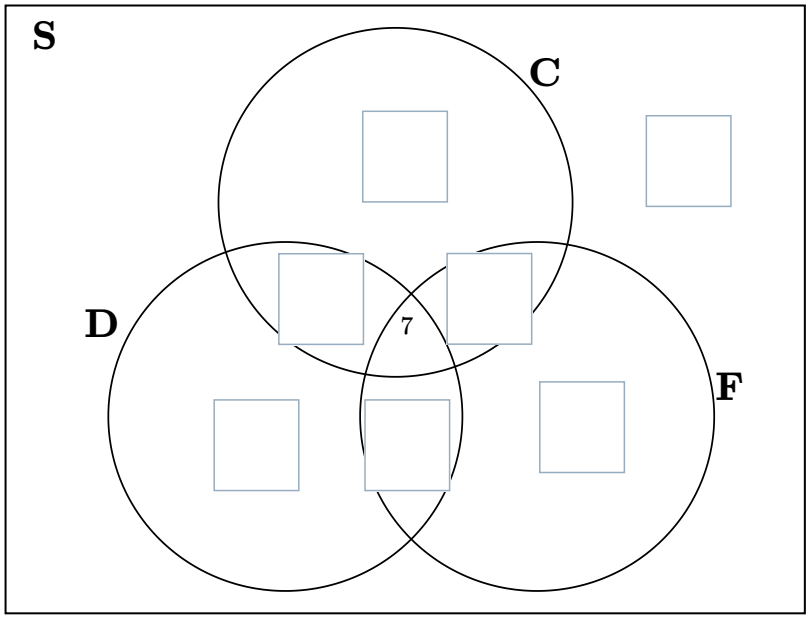
We can either put frequencies or probabilities into the Venn Diagram.

Given that $P(A) = 0.6$ and $P(A or B) = 0.85$, find the probability of:
a) $P(not A and B)$
b) $P(neither A nor B)$

Example involving frequencies 12

A vet surveys 100 of her clients. She finds that
25 own dogs, 15 own dogs and cats, 11 own dogs and tropical fish, 53 own cats, 10 own cats and tropical fish, 7 own dogs, cats and tropical fish, 40 own tropical fish.

Fill in this Venn Diagram, and hence answer the following questions:
a) $P(owns dog only)$
b) $P(does not own tropical fish)$
c) $P(does not own dogs, \ \ cats, or tropical fish)$



Tip: Start from the centre frequency and work your way outwards using subtraction.

a

b

c

Test Your Understanding

Jan 2012 Q6. Of 100 people, 65 run, 48 swim, 60 cycle, 40 run and swim, 30 swim and cycle, 35 run and cycle and 25 do all three.

(a) Draw a Venn diagram. (4 marks) Tip: include the box.

Find the probability that a randomly chosen person:

(b) takes none of these types of exercise (2 marks);

(c) swims but does not run (2 marks);

(d) takes at least two types of exercise (2 marks).

(e) Given that Jason runs, find the probability that he swims but does not cycle. (3 marks)

a

b

c

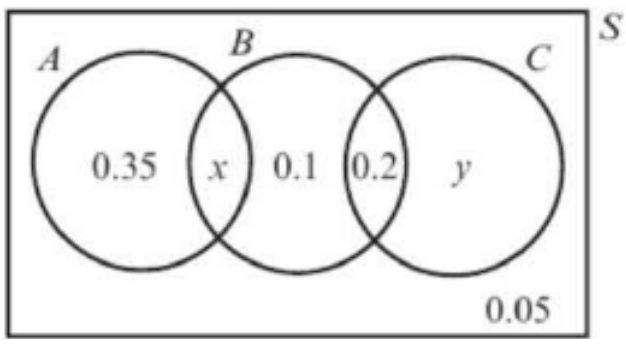
d

e

Exercise 5B – Statistics Year 1 page 74-75

5. In a doctor's waiting room a patient reads Hiya with probability 0.6, Dakar with probability 0.4, and either or both with probability 0.7. Find the probability of reading (a) both magazines (2 marks), (b) Hiya only (2 marks).

6. The diagram shows participation in archery (A), badminton (B) and croquet (C). Given $P(B) = 0.45$, find (a) x (1 mark) and (b) y (2 marks).



5a

5b

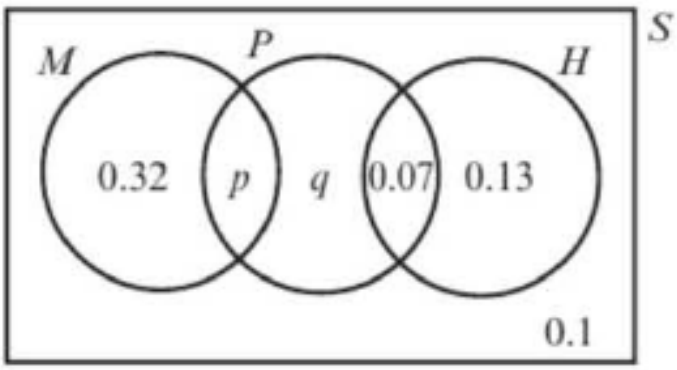
6a

6b

Exercise 5B – Statistics Year 1 page 74-75 15

7. The diagram shows which subjects sixth-form students study. M is mathematics, P physics, H history.

Given $P(M) = P(P)$, find p and q. (4 marks)

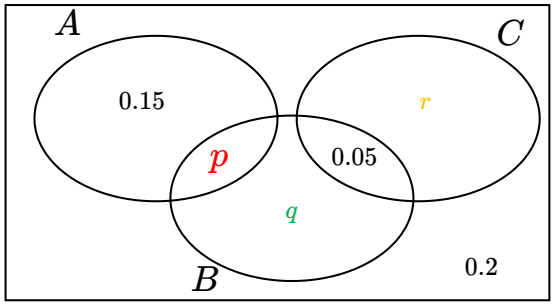


7

Extension 16

[Challenge p75] The Venn Diagram shows the probabilities of a group of children liking three types of sweet.

Given that $\mathrm{P}(B) = 2\mathrm{P}(A)$ and that $\mathrm{P}(notC) = 0.83$, find the values of p, q, and r.



Mutually Exclusive Events 17

If two events are mutually exclusive,

If A and B are mutually exclusive, then

$P(A \cap B) =$

$P(A \cup B) =$

The Venn diagram would look like this.

Mutually exclusive events have no outcomes in common. Their regions cannot overlap.

Independent Events 18

If two events are independent,

If A and B are independent, then

$P(A \cap B) =$

Note: Independence does not affect how the circles interact in a Venn Diagram.

Example 1 2 3 4

1 I pick one of 1, 2, 3, 4 at random. What is the probability that:

I pick a multiple of 2?

I pick a multiple of 4?

2 Explain conceptually why the events are not independent.

3 Show that the events are not independent.

This is a common exam question. Either show that $P(AandB) = P(A) \times P(B)$
or that $P(AandB) \neq P(A) \times P(B)$

Further Examples 19

[Textbook] Events A and B are mutually exclusive and $P(A) = 0.2$ and $P(B) = 0.4$.

a) Find $P(A or B)$

b) Find $P(A but not B)$

c) Find $P(neither A nor B)$

a

b

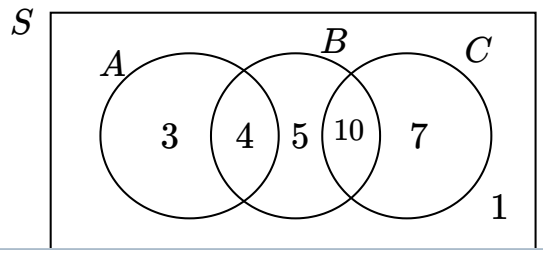
c

[Textbook] Events A and B are independent and $P(A) = \frac{1}{3}$ and $P(B) = \frac{1}{5}$. Find $P(A and B)$.

[Textbook] The Venn diagram shows the number of students in a particular class who watch any of three popular TV programmes.

a) Find the probability that a student chosen at random watches B or C or both.

b) Determine whether watching A and watching B are statistically independent.



a

b

Test Your Understanding 20

There are three events A, B, C. The events A and B are mutually exclusive.

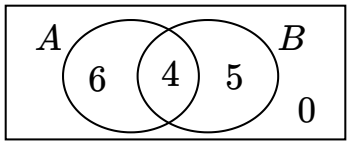
a) Draw a Venn diagram which represents this information.

b) If $P(A) = 0.1$ and $P(B) = 0.6$, determine $P(neither A nor B)$

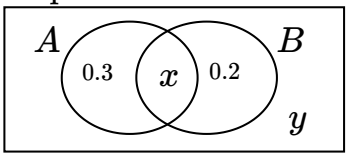
a

b

The Venn diagram shows the number of people who like each of two different colours. Determine if A and B are independent.



The Venn diagram shows the probability of each event. Given that A and B are independent, determine the possible values of x.

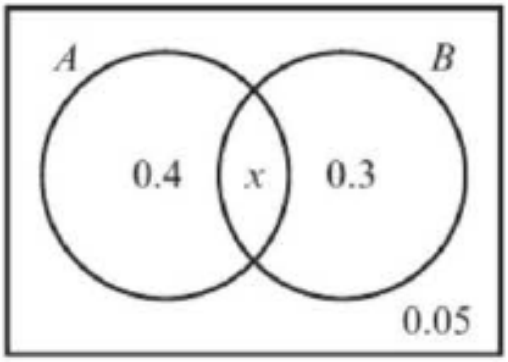


6. The Venn diagram shows the probabilities that students like pasta (A) or pizza (B).

(a) Write down the value of x. (1 mark)

(b) Determine whether liking pasta and liking pizza are independent. (3 marks)

8. $P(W) = 0.5$, $P(W \cap X') = 0.25$ and $P((W \cup X)') = 0.3$. Are W and X independent? Give a reason. (3 marks)



6a

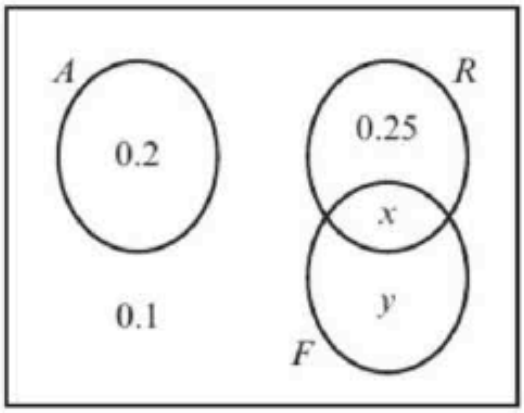
6b

8

9. A social club runs an archery competition (A), raffle (R) and fun run (F). The Venn diagram shows members' participation. The probability of taking part in archery or the raffle is 0.6.

(a) Find x and y. (2 marks)

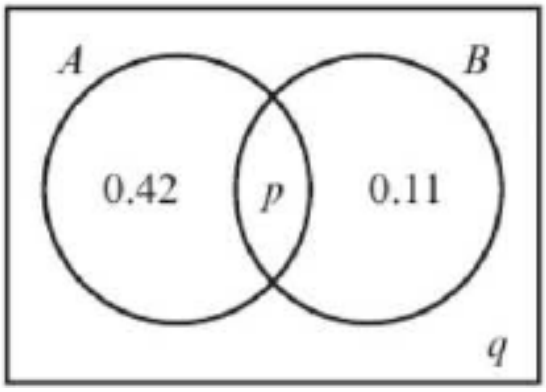
(b) Show that R and F are not independent. (3 marks)



a

b

10. Given that A and B are independent, find the two possible pairs of values for p and q.



10

Extension 24

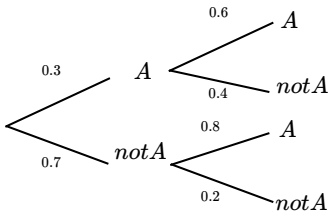
If A and B are independent, prove that:
(a) A and the complement of B are independent;
(b) the complements of A and B are independent.

a

b

At GCSE we saw that tree diagrams were an effective way of showing the outcome of two events which happen **in succession**.

(Personal opinion however is that their use is easily avoidable)



There are 3 yellow and 2 green counters in a bag. I take two counters at random. Determine the probability that:

a) They are of the same colour.

b) They are of different colours.

a

b

list out the matching sequences of outcomes first, then find the probability for each.

Note that if the outcomes are the same but reordered, the probability will be the same. So we could have quickly done $\frac{2}{5} \times \frac{3}{4} \times 2$

The probability I hit a target on each shot is 0.3. I keep firing until I hit the target. Determine the probability I hit the target on the 5th shot.

4. Mr Dixon scores par or under on the first hole with probability 0.7. Given par or under on the first hole, the probability of par or under on the second is 0.8. Otherwise it is 0.4.

(a) Draw a tree diagram. (3 marks)

(b) Are scoring par or under on each of the two holes independent events? (1 mark)

(c) Find the probability of par or under on exactly one hole. (3 marks)

b

c

a

5. A biased coin with $P(H) = 1/3$ is tossed three times.
(a) Draw a tree diagram. (3 marks)
(b) Find the probability of three heads. (1 mark)
(c) Find the probability of exactly one head. (2 marks)
(d) Repeat the experiment for a second trial. Find the probability of obtaining three heads or three tails in both trials. (3 marks)

a

b

c

d

6. A bag contains 13 tokens: 4 blue, 3 red and 6 yellow. Draw two without replacement.
(a) Find the probability that both are yellow. (2 marks)
Draw a third token.
(b) Find the probability it is yellow given that the first two were yellow. (1 mark)
(c) Find the probability all three tokens are different colours. (4 marks)

a

b

c